Jérôme Botoko Ekila

Academic CV

Pleinlaan 9, 1050 Brussels, Belgium
jerome@ai.vub.ac.be

Education

Vrije Universiteit Brussel *Brussels, Belgium*
PhD in Sciences Nov. 2021 - present
- Promotors: Prof. dr. Paul Van Eecke & Prof. dr. Katrien Beuls
- Research on emergent communication and construction grammar

Vrije Universiteit Brussel *Brussels, Belgium*
Master of Science in Applied Sciences & Engineering: Computer Science Sep. 2018 - Sep. 2020
- Honor: magna cum laude (84%)
- Profile: Artificial Intelligence
- Thesis: Neuro-symbolic Procedural Semantics for Visual Question Answering

Vrije Universiteit Brussel *Brussels, Belgium*
Bachelor of Science: Computer Science Sep. 2016 - Sep. 2019
- Honor: magna cum laude (78%)

Research Experience

Sorbonne Université - Institute of Intelligent Systems and Robotics *Paris, France*
Visiting Researcher Nov. 2025 - Jan. 2026
- Advisors: Prof. dr. Nicolas Bredeche
- Research on emergent communication and swarm robotics

Vrije Universiteit Brussel - Artificial Intelligence Laboratory *Brussels, Belgium*
Research Engineer Nov. 2020 - Oct. 2021
- Developed two web applications demonstrating novel research from the EHAI research group
- Developed a neuro-symbolic AI system integrating linguistic and visual reasoning
- Assisted PhD researchers by analyzing and reproducing novel state-of-the-art methodologies

Publications

Journals

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2026). Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. *Transactions of the Association for Computational Linguistics (TACL)* 14 (pp. 2234-2257) doi: 10.1162/TACL.a.799 (Q1, Impact Factor: 11.7).

Wright, J., Verheyen, L., De Ponte, F., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (2026). Conventionality Produces More Robust, Learnable and Efficient Emergent Languages. *Journal of Language Evolution* 11 doi: 10.1093/jole/lzag006 (Q1, Impact Factor: 1.6).

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2025). Neuro-Symbolic Procedural Semantics for Explainable Visual Dialogue. *PLOS One* 20(5) doi: 10.1371/journal.pone.0323098 (Q1, Impact Factor: 3.3).

Van Eecke, P., Beuls, K., **Botoko Ekila, J.** & Rădulescu, R. (2022). Language games meet multi-agent reinforcement learning: A case study for the naming game. *Journal of Language Evolution* 7(2) (pp. 213-223) doi: 10.1093/jole/lzad001 (Q1, Impact Factor: 1.6).

Conferences

Toquebiau, M., **Botoko Ekila, J.**, Van Eecke, P. & Beuls, K. (2026). Cultural Evolution Drives the Emergence of Conventions in Neural Emergent Communication. *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP 2026)* (pp. 32202–32223) doi: 10.18653/2026.emnlp-main.1410 (A* conference, 15.4% acceptance).

Botoko Ekila, J., Nevens, J., Verheyen, L., Beuls, K. & Van Eecke, P. (2024). Decentralised Emergence of Robust and Adaptive Linguistic Conventions in Populations of Autonomous Agents Grounded in Continuous Worlds. *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024)* (pp. 2168-2170) (A conference, 45.1% acceptance).

Botoko Ekila, J.. (2024). Emergence of Linguistic Conventions in Multi-Agent Systems Through Situated Communicative Interactions. *Proceedings of the 23rd International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2024)* (pp. 2725–2727) (A conference, 45.1% acceptance).

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2023). Neuro-Symbolic Procedural Semantics for Reasoning-Intensive Visual Dialogue Tasks. *26th European Conference on Artificial Intelligence (ECAI 2023)* (pp. 2419-2426) doi: 10.3233/FAIA230544 (A conference, 23.9% acceptance).

Workshops

Botoko Ekila, J., Verheyen, L., Beuls, K. & Van Eecke, P. (2025). Constructions All the Way Up: From Sensory Experiences to Construction Grammars. *Proceedings of the Second International Workshop on Construction Grammars and NLP (CxGs+NLP 2025)* (pp. 84-95).

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2022). Hybrid Procedural Semantics for Visual Dialogue: An Interactive Web Demonstration. In *Proceedings of the 31st International Joint Conference on Artificial Intelligence (IJCAI 2022): Workshop on semantic techniques for narrative-based understanding* (pp. 48-52).

Peer-reviewed abstracts (Presentations)

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2026). Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. TACL paper accepted for oral presentation (presented by Jérôme Botoko Ekila) *Empirical Methods in Natural Language Processing (EMNLP 2026), Budapest, Hungary.*

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2026). Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. Encore abstract accepted for oral presentation (presented by Jérôme Botoko Ekila) *Belgium-Netherlands Conference on Artificial Intelligence (BNAIC/BENELEARN 2026), Maastricht, Netherlands.*

Toquebiau M., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (2026). Cultural Evolution Drives the Emergence of Conventions in Neural Emergent Communication. Abstract accepted for oral presentation (presented by Maxime Toquebiau). *Computational Linguistics in the Netherlands 36 (CLIN 36), Brussels, Belgium.*

Jamie Wright, Verheyen, L., De Ponte, F., **Botoko Ekila, J.**, Beuls, K. & Van Eecke, P. (2025). Conventionality Produces More Robust, Learnable and Efficient Emergent Languages. Abstract accepted as poster. *Computational Linguistics in the Netherlands 35, Leuven, Belgium.*

Botoko Ekila, J., Verheyen, L., Beuls, K. & Van Eecke, P. (2025). Constructions All the Way Up: From Sensory Experiences to Construction Grammars. Abstract accepted as poster. *Computational Linguistics in the Netherlands 35 (CLIN 35), Leuven, Belgium.*

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2024). Decentralised Emergence of Robust and Adaptive Linguistic Conventions in Populations of Autonomous Agents Grounded in Continuous Worlds. Accepted for oral presentation (presented by Jérôme Botoko Ekila). *Adaptive and Learning Agents Workshop @ AAMAS (ALA 2024), Auckland, New-Zealand.*

Van Eecke, P., Beuls, K., **Botoko Ekila, J.** & Rădulescu, R. (2023). Investigating emergent communication through language games and multi-agent reinforcement learning. Encore abstract accepted for oral presentation (presented by Jérôme Botoko Ekila). *Belgium-Netherlands Conference on Artificial Intelligence (BNAIC/BENELEARN 2023), Delft, Netherlands.*

Botoko Ekila, J., Verheyen, L., Nevens, J., Beuls, K. & Van Eecke, P. (2023). Decentralised Emergence of Robust and Adaptive Linguistic Conventions in Populations of Autonomous Agents Grounded in Continuous Worlds. Abstract accepted as poster. *Computational Linguistics in the Netherlands 33 (CLIN 33), Antwerp, Belgium.*

Verheyen, L., **Botoko Ekila, J.**, Nevens, J., Van Eecke, P. & Beuls, K. (2022). Hybrid Procedural Semantics. Abstract accepted for oral presentation (presented by Lara Verheyen). *Computational Linguistics in the Netherlands 32 (CLIN 32), Tilburg, Netherlands.*

Van Eecke, P., Beuls, K., **Botoko Ekila, J.** & Rădulescu, R. (2022). Investigating emergent communication through language games and multi-agent reinforcement learning. Abstract accepted as poster. *Computational Linguistics in the Netherlands 32 (CLIN 32), Tilburg, Netherlands.*

Invited Talks and Guest Lectures

Oct. 2026. Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. **Guest lecture for the Natural Language Processing course**, Université de Namur, Namur, Belgium

Mar. 2025. Emergent Communication in Continuous Worlds: Self-Organisation of Conceptually Grounded Vocabularies at Scale. **Invited talk at Nicolas Bredeche's lab, Institute for Intelligent Systems and Robotics (ISIR)**, Sorbonne Université, Paris, France.

Media Coverage

Trends (Belgian online article), 14 October 2026. "Belgian university research shows how AI agents develop their own natural language to communicate".

Trends-Tendances (Belgian magazine), 31 October 2026. "In this Belgian Lab AI Invents Its Own Language".

Teaching Experience

2026-2027	**Course: Natural Language Processing** Teaching Assistant, 6 ECTS	*Prof. dr. Paul Van Eecke*
2023-2027	**Course: Introduction to Computational Linguistics** Teaching Assistant, 3 ECTS	*Prof. dr. Paul Van Eecke*
2023-2027	**Course: Advanced Topics in Natural Language Processing** Teaching Assistant, 6 ECTS	*Prof. dr. Paul Van Eecke*
2022-2027	**Course: Wetenschappelijk Rekenen** Teaching Assistant, 6 ECTS	*Prof. dr. Kurt Barbé, Prof. dr. Ann Nowé*
2024	**Course: Artificial Intelligence Paradigms** Teaching Assistant, 6 ECTS	*Prof. dr. Paul Van Eecke*
2022	**Course: Artificial Intelligence** Teaching Assistant, 6 ECTS	*Prof. dr. dr. Geraint Wiggins*

Academic Service and Community Engagement

Reviewing

2025-2026	**Benelux Conference on Artificial Intelligence (BNAIC)**
2024-2026	**Adaptive and Learning Agents - Workshop at AAMAS**
2025	**International Workshop on Construction Grammars and NLP (CxGs+NLP)**
2023	**European Conference on Artificial Intelligence (ECAI)**

Service

2026	**Organising committee - Evolutionary Linguistics Association (ELA) Symposium**	*Paris, France*
2026	**Organising committee - Computational Linguistics in the Netherlands (CLIN36)**	*Brussels, Belgium*
2023	**Volunteer - Belgium NLP Meetup 2023**	*Brussels, Belgium*

Awards, Fellowships, & Grants

July 2025	**Mobility Grant for long stay abroad (3 months)** Research Foundation Flanders (FWO)
July 2025	**Mobility Grant for participation in a workshop** Research Foundation Flanders (FWO)
May 2024	**Student Scholarship AAMAS** International Foundation for Autonomous Agents and Multiagent Systems (IFAAMAS)

Mentoring

June 2026	**Jitse De Beer** Bachelor Thesis - *Pre-linguistic Category Formation in Grounded Language Games*, Vrije Universiteit Brussel
June 2026	**Farah Ayyad** Master Thesis - *Concept Representation and Lexical Convergence in Grounded Language Games*, Vrije Universiteit Brussel
Sep. 2025	**Xantin Préser** Master Thesis - *Grounded concepts in multi-word utterances: A methodology for the emergence of compositional languages*, Vrije Universiteit Brussel & UNamur
June 2025	**Maël-Questiaux** Master Thesis - *Language games in Virtual Reality*, Vrije Universiteit Brussel & UNamur
Sep. 2024	**Khalil Ouald Chaib** Master thesis - *LLM Detection: PCFG-based authorship attribution for detecting LLM-generated texts*, Vrije Universiteit Brussel